

DAX CALCULATIONS

1. Add a calculated column in the order details that combines the “category” and “sub-category” columns into a single ‘category type’ column.

The screenshot shows the Power BI Desktop interface. The formula bar at the top displays the DAX formula for the 'Category type' column: `1 Category type = 'Order Details'[Category]&" "&'Order Details'[Sub-Category]`. Below the formula bar, a table of data is shown with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, Profit Status, Revenue, sales category, and Category type. The 'Category type' column contains the concatenated values of 'Category' and 'Sub-Category' for each row. The right-hand pane shows the 'Data' view with a search bar and a list of fields including 'Category type'.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Revenue	sales category	Category type
B-25604	65	17	2	Clothing	T-shirt	0.2615	Profit	130	Below Average	ClothingT-shirt
B-25606	87	4	2	Clothing	Shirt	0.046	Profit	174	Below Average	ClothingShirt
B-25607	50	15	4	Clothing	Leggings	0.3	Profit	200	Below Average	ClothingLeggings
B-25615	68	20	5	Clothing	Hankercchief	0.2941	Profit	340	Below Average	ClothingHankercchief
B-25616	42	12	5	Clothing	Hankercchief	0.2857	Profit	210	Below Average	ClothingHankercchief
B-25619	353	90	8	Clothing	Saree	0.255	Profit	2824	Above Average	ClothingSaree
B-25623	53	1	4	Clothing	Stole	0.0189	Profit	212	Below Average	ClothingStole
B-25624	26	12	3	Clothing	Hankercchief	0.4615	Profit	78	Below Average	ClothingHankercchief
B-25625	97	29	2	Clothing	Hankercchief	0.299	Profit	194	Below Average	ClothingHankercchief
B-25628	45	13	4	Clothing	Skirt	0.2889	Profit	180	Below Average	ClothingSkirt
B-25629	1560	421	3	Clothing	Trousers	0.2699	Profit	4680	Above Average	ClothingTrousers
B-25630	133	12	5	Clothing	Stole	0.0902	Profit	665	Below Average	ClothingStole
B-25635	40	16	3	Clothing	Hankercchief	0.4	Profit	120	Below Average	ClothingHankercchief
B-25636	637	113	5	Clothing	Saree	0.1774	Profit	3185	Above Average	ClothingSaree
B-25637	117	14	3	Clothing	Shirt	0.1197	Profit	351	Below Average	ClothingShirt
B-25642	434	26	11	Clothing	Shirt	0.0599	Profit	4774	Above Average	ClothingShirt
B-25654	34	12	3	Clothing	Hankercchief	0.3529	Profit	102	Below Average	ClothingHankercchief
B-25658	27	9	2	Clothing	T-shirt	0.3333	Profit	54	Below Average	ClothingT-shirt
B-25659	148	72	7	Clothing	T-shirt	0.4865	Profit	1036	Below Average	ClothingT-shirt
B-25670	74	29	3	Clothing	Stole	0.3919	Profit	222	Below Average	ClothingStole
B-25687	17	6	1	Clothing	Stole	0.3529	Profit	17	Below Average	ClothingStole
B-25690	31	10	3	Clothing	Skirt	0.3226	Profit	93	Below Average	ClothingSkirt
B-25691	714	56	4	Clothing	Saree	0.0784	Profit	2856	Above Average	ClothingSaree
B-25694	167	43	7	Clothing	T-shirt	0.2575	Profit	1169	Below Average	ClothingT-shirt
B-25695	171	14	9	Clothing	Shirt	0.0819	Profit	1539	Below Average	ClothingShirt
B-25717	561	212	3	Clothing	Saree	0.3779	Profit	1683	Above Average	ClothingSaree
B-25719	29	10	2	Clothing	Stole	0.3448	Profit	58	Below Average	ClothingStole

2. create a calculate column in the order table to compute the revenue (Amount*Quantity) per order.

The screenshot shows the Power BI Desktop interface. The formula bar at the top displays the DAX formula for the 'Revenue' column: `1 Revenue = ('Order Details'[Amount] * 'Order Details'[Quantity])`. Below the formula bar, a table of data is shown with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, Profit Status, Revenue, sales category, and Category type. The 'Revenue' column contains the calculated values of 'Amount' multiplied by 'Quantity' for each row. The right-hand pane shows the 'Data' view with a search bar and a list of fields including 'Revenue'.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Revenue	sales category	Category type
B-25604	65	17	2	Clothing	T-shirt	0.2615	Profit	130	Below Average	ClothingT-shirt
B-25606	87	4	2	Clothing	Shirt	0.046	Profit	174	Below Average	ClothingShirt
B-25607	50	15	4	Clothing	Leggings	0.3	Profit	200	Below Average	ClothingLeggings
B-25615	68	20	5	Clothing	Hankercchief	0.2941	Profit	340	Below Average	ClothingHankercchief
B-25616	42	12	5	Clothing	Hankercchief	0.2857	Profit	210	Below Average	ClothingHankercchief
B-25619	353	90	8	Clothing	Saree	0.255	Profit	2824	Above Average	ClothingSaree
B-25623	53	1	4	Clothing	Stole	0.0189	Profit	212	Below Average	ClothingStole
B-25624	26	12	3	Clothing	Hankercchief	0.4615	Profit	78	Below Average	ClothingHankercchief
B-25625	97	29	2	Clothing	Hankercchief	0.299	Profit	194	Below Average	ClothingHankercchief
B-25628	45	13	4	Clothing	Skirt	0.2889	Profit	180	Below Average	ClothingSkirt
B-25629	1560	421	3	Clothing	Trousers	0.2699	Profit	4680	Above Average	ClothingTrousers
B-25630	133	12	5	Clothing	Stole	0.0902	Profit	665	Below Average	ClothingStole
B-25635	40	16	3	Clothing	Hankercchief	0.4	Profit	120	Below Average	ClothingHankercchief
B-25636	637	113	5	Clothing	Saree	0.1774	Profit	3185	Above Average	ClothingSaree
B-25637	117	14	3	Clothing	Shirt	0.1197	Profit	351	Below Average	ClothingShirt
B-25642	434	26	11	Clothing	Shirt	0.0599	Profit	4774	Above Average	ClothingShirt
B-25654	34	12	3	Clothing	Hankercchief	0.3529	Profit	102	Below Average	ClothingHankercchief
B-25658	27	9	2	Clothing	T-shirt	0.3333	Profit	54	Below Average	ClothingT-shirt
B-25659	148	72	7	Clothing	T-shirt	0.4865	Profit	1036	Below Average	ClothingT-shirt
B-25670	74	29	3	Clothing	Stole	0.3919	Profit	222	Below Average	ClothingStole
B-25687	17	6	1	Clothing	Stole	0.3529	Profit	17	Below Average	ClothingStole
B-25690	31	10	3	Clothing	Skirt	0.3226	Profit	93	Below Average	ClothingSkirt
B-25691	714	56	4	Clothing	Saree	0.0784	Profit	2856	Above Average	ClothingSaree
B-25694	167	43	7	Clothing	T-shirt	0.2575	Profit	1169	Below Average	ClothingT-shirt
B-25695	171	14	9	Clothing	Shirt	0.0819	Profit	1539	Below Average	ClothingShirt
B-25717	561	212	3	Clothing	Saree	0.3779	Profit	1683	Above Average	ClothingSaree
B-25719	29	10	2	Clothing	Stole	0.3448	Profit	58	Below Average	ClothingStole

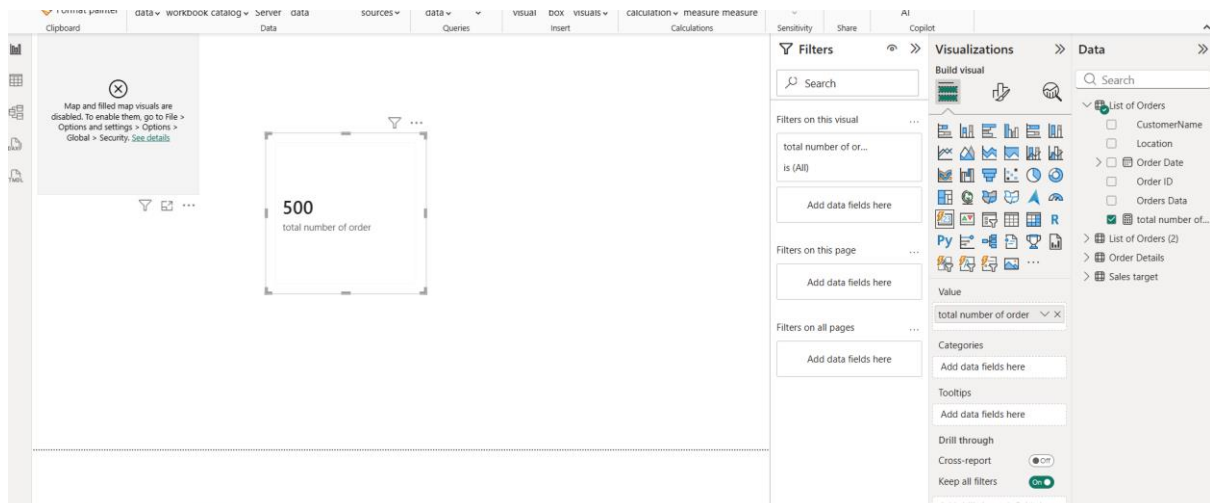
3. Add a calculate column named ‘sales category’ in the order details that categories each order as ‘ Above average’ or below average on the amount values.

1 sales category = if('Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Average","Below Average")

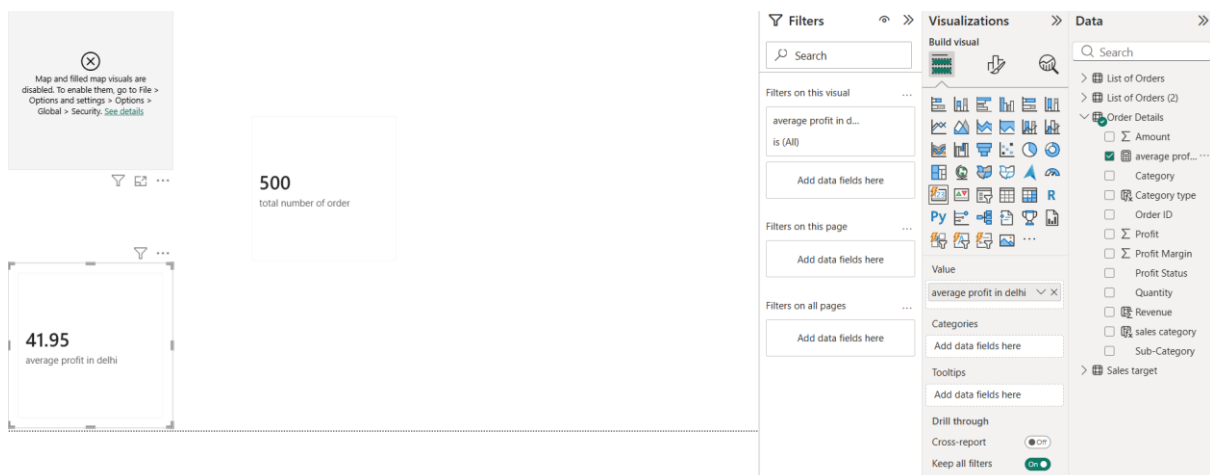
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Revenue	sales category
B-25604	65	17	2	Clothing	T-shirt	0.2615	Profit	130	Below Average
B-25606	87	4	2	Clothing	Shirt	0.046	Profit	174	Below Average
B-25607	50	15	4	Clothing	Leggings	0.3	Profit	200	Below Average
B-25615	68	20	5	Clothing	Hankerschief	0.2941	Profit	340	Below Average
B-25616	42	12	5	Clothing	Hankerschief	0.2857	Profit	210	Below Average
B-25619	353	90	8	Clothing	Saree	0.255	Profit	2824	Above Average
B-25623	53	1	4	Clothing	Stole	0.0189	Profit	212	Below Average
B-25624	26	12	3	Clothing	Hankerschief	0.4615	Profit	78	Below Average
B-25625	97	29	2	Clothing	Hankerschief	0.299	Profit	194	Below Average
B-25628	45	13	4	Clothing	Skirt	0.2889	Profit	180	Below Average
B-25629	1560	421	3	Clothing	Trousers	0.2699	Profit	4680	Above Average
B-25630	133	12	5	Clothing	Stole	0.0902	Profit	665	Below Average
B-25635	40	16	3	Clothing	Hankerschief	0.4	Profit	120	Below Average
B-25636	637	113	5	Clothing	Saree	0.1774	Profit	3185	Above Average
B-25637	117	14	3	Clothing	Shirt	0.1197	Profit	351	Below Average
B-25642	434	26	11	Clothing	Shirt	0.0599	Profit	4774	Above Average
B-25654	34	12	3	Clothing	Hankerschief	0.3529	Profit	102	Below Average
B-25658	27	9	2	Clothing	T-shirt	0.3333	Profit	54	Below Average
B-25659	148	72	7	Clothing	T-shirt	0.4865	Profit	1036	Below Average
B-25670	74	29	3	Clothing	Stole	0.3919	Profit	222	Below Average
B-25687	17	6	1	Clothing	Stole	0.3529	Profit	17	Below Average
B-25690	31	10	3	Clothing	Skirt	0.3226	Profit	93	Below Average
B-25691	714	56	4	Clothing	Saree	0.0794	Profit	2856	Above Average
B-25694	167	43	7	Clothing	T-shirt	0.2575	Profit	1169	Below Average
B-25695	171	14	9	Clothing	Shirt	0.0819	Profit	1539	Below Average
B-25717	561	212	3	Clothing	Saree	0.3779	Profit	1683	Above Average
B-25719	29	10	2	Clothing	Stole	0.3448	Profit	58	Below Average
B-25746	87	16	3	Clothing	Saree	0.1820	Profit	174	Below Average

Measures

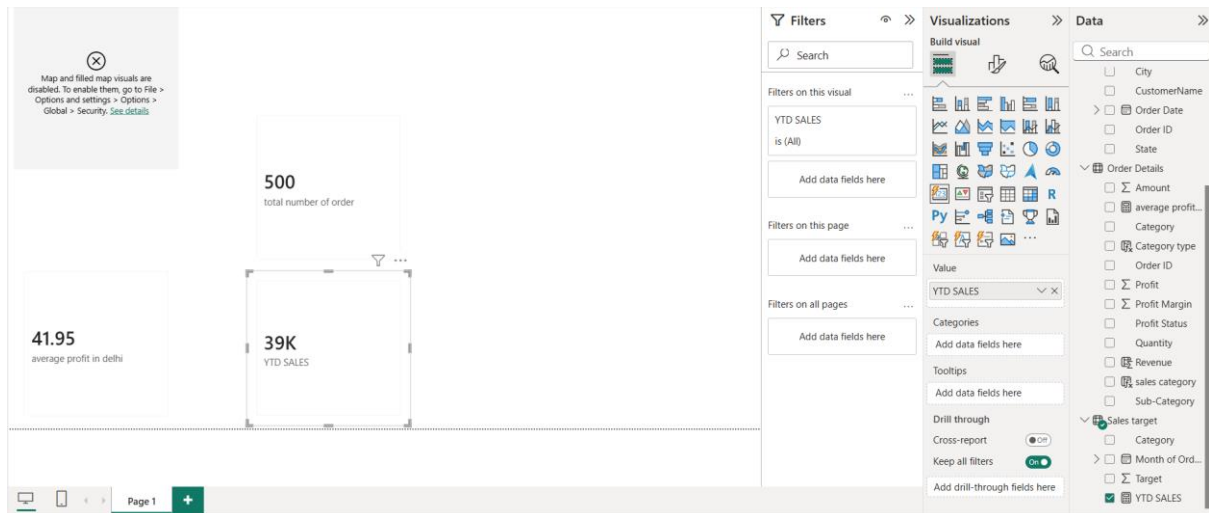
1.create a measure that calculate the total number of orders in the order details table.



2. Average profit in Delhi – create a measures that calculates the average profit for orders placed in Delhi.

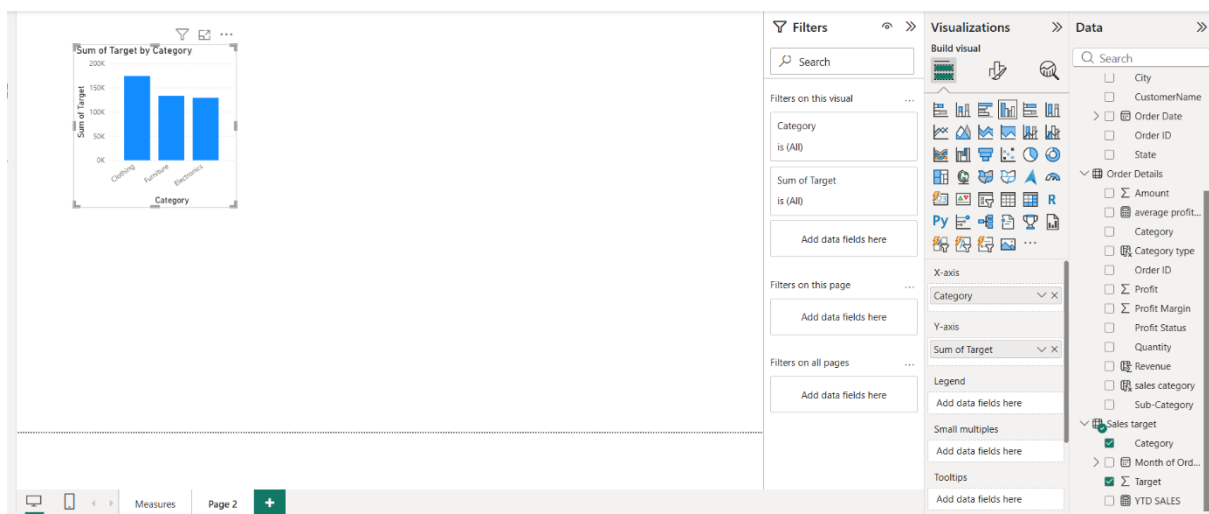


3. year – to – date (YTD) sales- create a measures that calculates total sales accumulated from the beginning of the year up to the current order date.

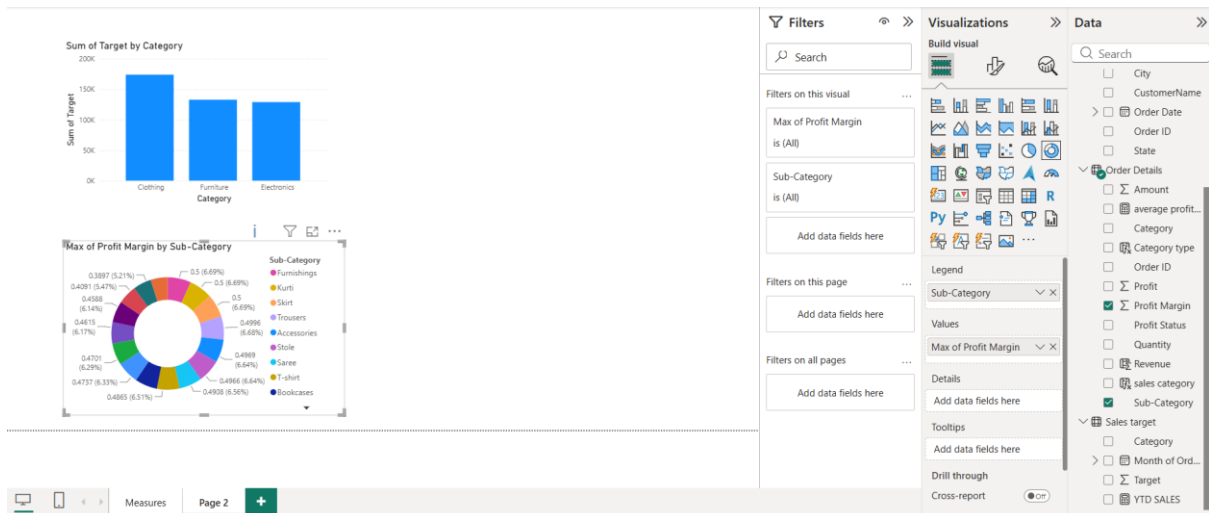


Data Visualization

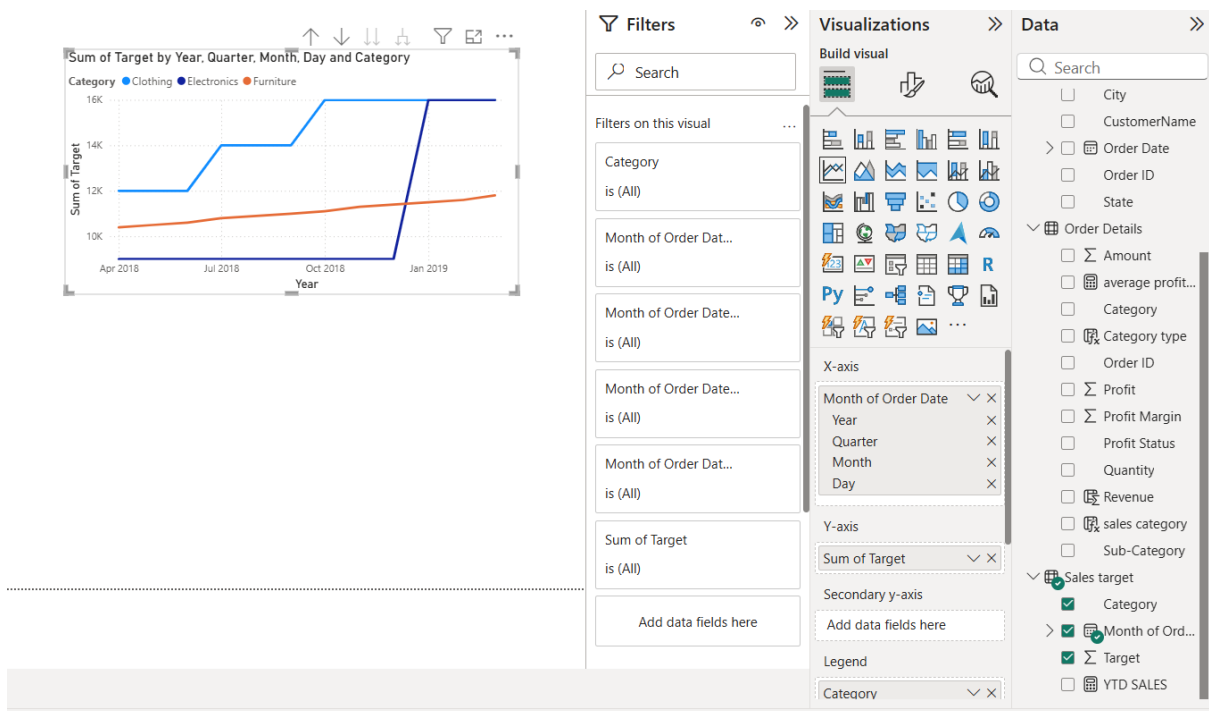
1. sales target achievement by category : compare actual sales with sales targets by category using a clustered column chart.



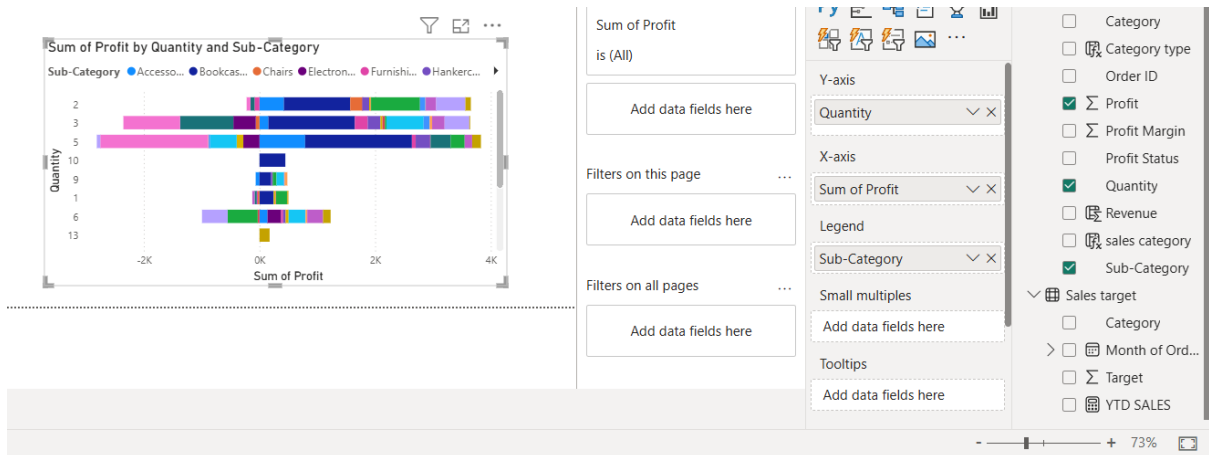
2. max profit margin by sub- category : Analyze the maximum profit margin for each sub – category of products using a donut chat.



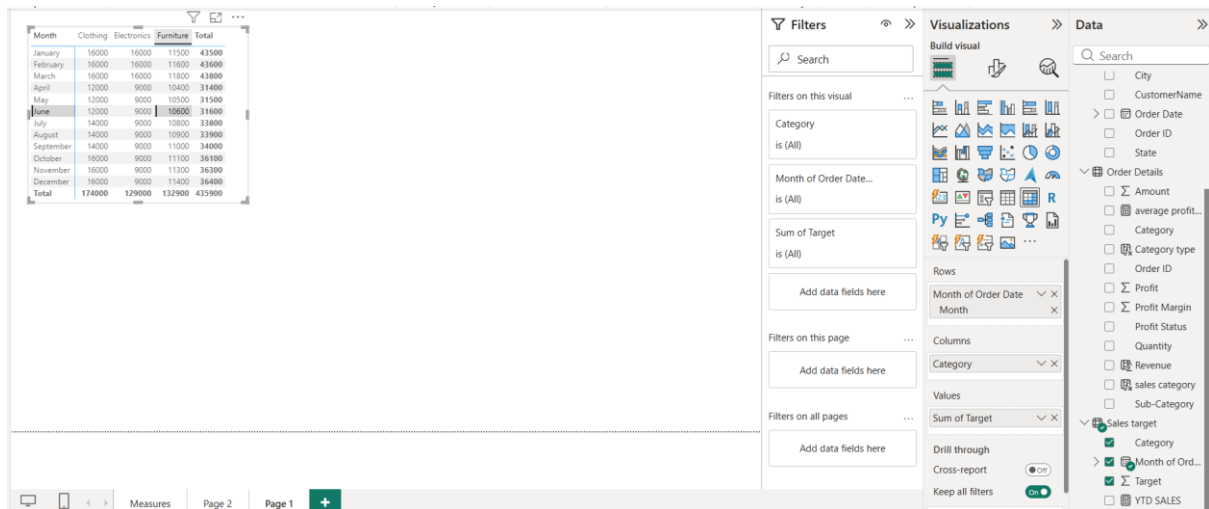
3. Monthly sales Trend : Show the trend of monthly sales over time using a line chat.



4. Comparison of profit and Quantity by sub -Category : Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



6. sales performance matrix : Built a matrix view to analyze how actual sales compare to sales targets across different category and months.



8.visualize total sales on a map by city to identify regional sales patters.

9. sub category using a tree map.

10. create a funnel chart to visualize the distribution of order counts across different states.

